

Investment Committee Memorandum

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TO	Investment Committee — Series B lead investor	DATE	September 2026
FROM	DY Research	ENGAGEMENT	DYR-DD-2026-009 · Due Diligence
RE	Meridane Robotics — the MR-7 joint controller	READING TIME	Four minutes

RECOMMENDATION

Proceed, with two conditions precedent.

The technology is real and well engineered. Two of its published claims do not hold as built. Both are fixable in software within weeks, and neither changes the thesis. One of them changes the safety timeline.

886 µs worst-case servo latency claimed: under 500	2,970 µs fault to stop claimed: under 2,000	488 µs after two contained software changes	0 findings that change the technical thesis
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What holds

- ✓ **The architecture.** Fixed-priority Rust firmware with no allocator on any control path. A sound base for a certifiable product.
- ✓ **The engineering.** Every push is built and tested; 31 hardware-in-the-loop tests run nightly; builds reproduce bit for bit.
- ✓ **The licences.** All 114 dependencies are permissive. Nothing copyleft ships in the binary.

What does not

- **"Latency under 500 µs."** 886 µs as built: a 180 µs telemetry lock blocks the servo loop. Its budget is 18 µs.
- **"Safe stop within 2 ms."** 2,970 µs: faults are polled every 2 ms. No priority change can fix a polling period.
- **The timing margin.** After the fixes, 12 µs, measured against bench maxima rather than proven bounds.

The five risks that matter

RISK	LIKELIHOOD	IMPACT	MITIGATION
Safety certification slips. The safety concept relies on a 2 ms stop the design cannot deliver.	HIGH	HIGH	Condition precedent 2.
A published claim fails in the field. The 500 μs figure is in the datasheet and in two customer contracts.	HIGH	MEDIUM	Condition precedent 1.
The margin is thinner than measured. 12 μs rests on a 210 μs execution time that is a bench maximum.	MEDIUM	MEDIUM	Condition subsequent 1.
Key people. Two engineers wrote 81% of the control firmware.	MEDIUM	HIGH	Retention terms; a documentation milestone.
A single encoder supplier. No second source is qualified.	LOW	MEDIUM	Qualify a second source within 12 months.

Conditions

CP 1 Telemetry off the shared lock. The servo loop's response under 500 μs at every phasing, shown on the bench with a GPIO test.	CP 2 Stop on an interrupt. Fault to gate-driver disable under 50 μs on a scope, and the safety concept revised to match.	CS 1 A bound, not a maximum. The servo task's execution time with a stated method, clearing 222 μs with margin, within 90 days of close.
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Effect on the thesis

None of the findings changes what the technology is or what it can become. Two change when it can ship under a safety certificate: we would plan six to eight more weeks on the safety track, and price that time into the close.

Where this memo stops

It summarises a three-week technical due diligence, and the full report with its evidence appendix. It is not investment advice: the decision, and the price, remain the committee's.

DY Research · dyresearch.github.io · connect@axonos.org

This is a sample memorandum. Every report we write belongs to its client and is published only with their written consent, so Meridane Robotics and every fact about it here are fictional. The analysis and the arithmetic are real, and match the sample Focused Audit figure for figure.